

DETECTION OF SEVERITY IN TRAFFIC ACCIDENTS

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Introduction and Objectives

Road safety is a critical public health challenge. This exploratory analysis focuses on identifying severity patterns in traffic accidents using real data from 2020.

- 🎯 Understand the factors that influence accident severity.
- 📈 Identify critical variables for future predictive modeling.
- 🛡️ Support evidence-based prevention strategies.

22.643

RECORDS ANALYZED

2020

BASE YEAR

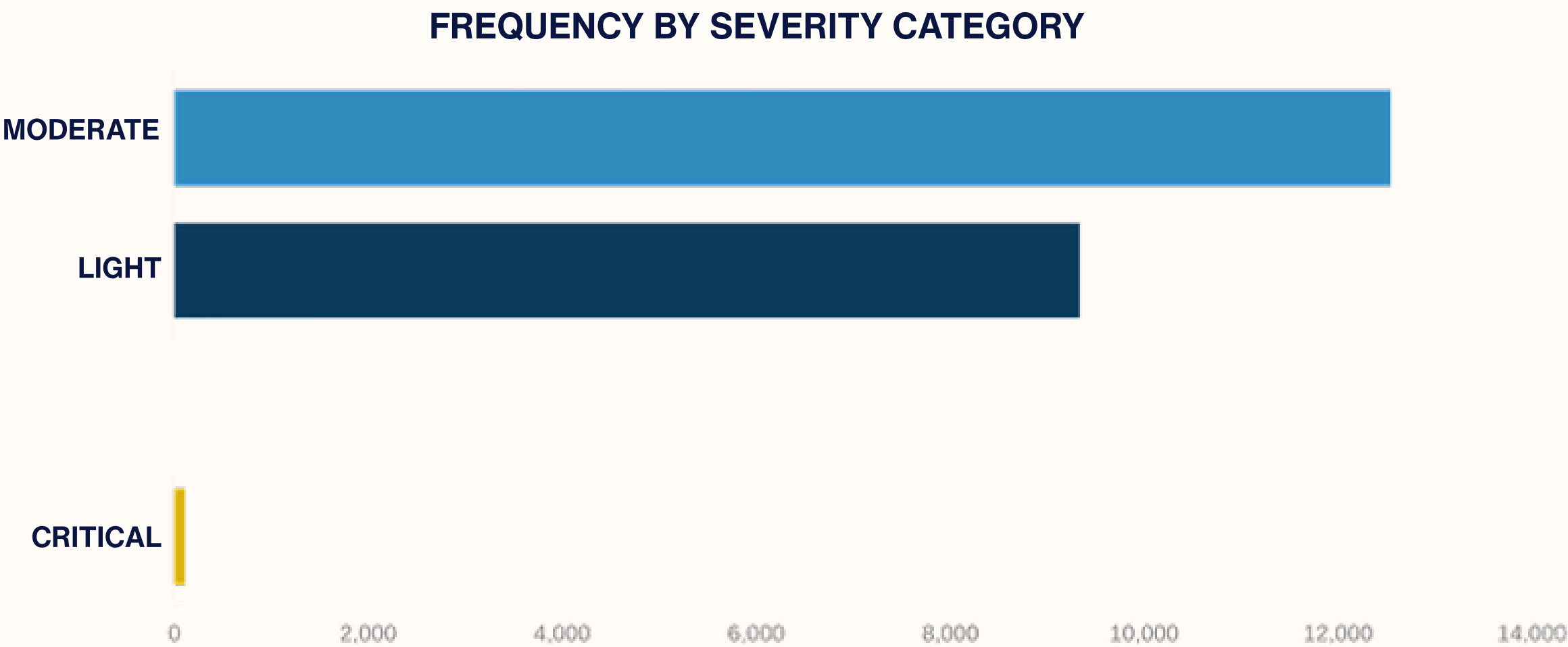
SEVERITY

TARGET VARIABLE

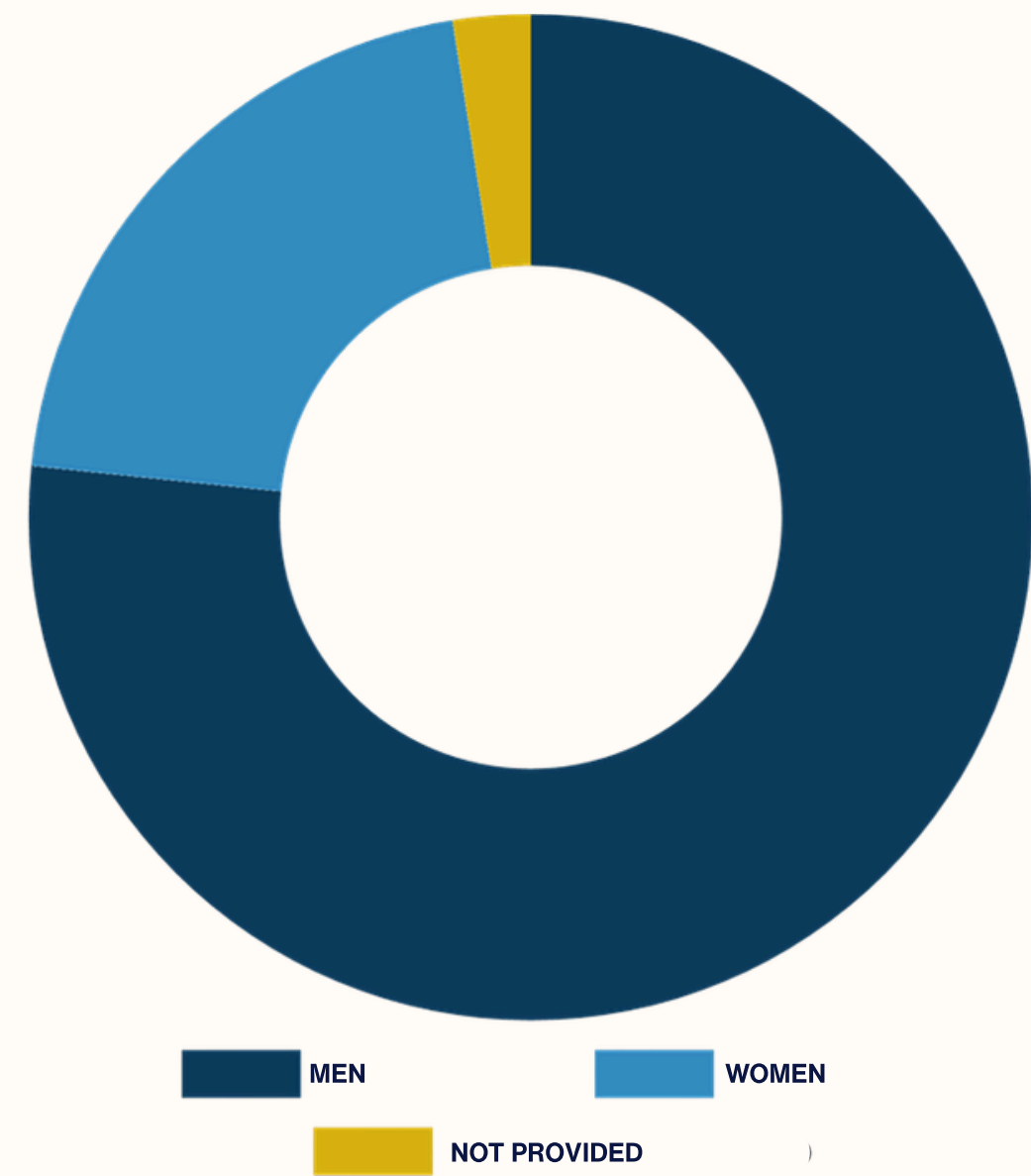


General Overview: Severity Distribution

The vast majority of occurrences result in non-fatal physical harm or property damage only. The small percentage of fatal cases (0.5%) represents the greatest challenge for public safety and predictive modeling.



Gender Distribution



Demographic Profile

♀♂ Gender Disparity

The data shows a marked male predominance (approx. 76%) across all severity categories, suggesting greater exposure or specific risk-taking behaviors.

17.351
MEN

4.727
WOMEN

Passive Safety: The Impact of Seatbelts

The vast majority of incidents result in non-lethal physical or purely material damage. The small percentage of fatal cases (0.5%) represents the greatest challenge for public safety and predictive modeling.

Seatbelt use is the single most decisive factor in mitigating injury severity in traffic accidents.

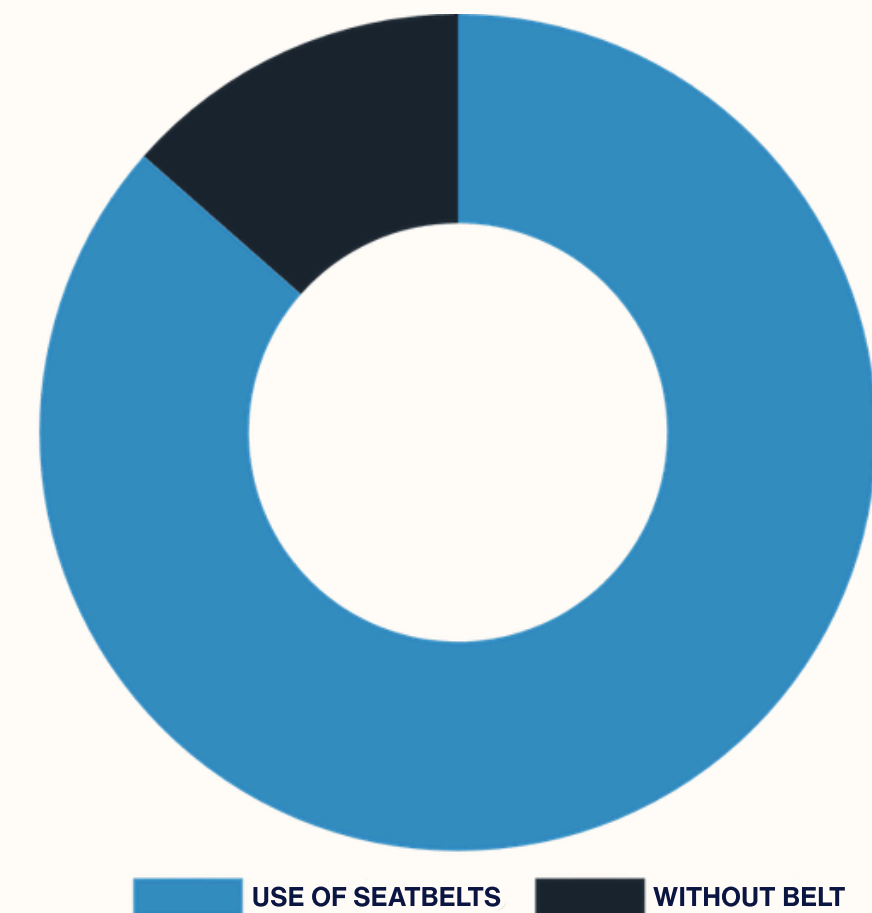
The absence of this device is correlated with higher-severity cases and disabling injuries.

19.583

involved were
wearing a
seatbelt (86.5%)

⚠️ 3.060

involved were not
wearing a
seatbelt (13.5%)

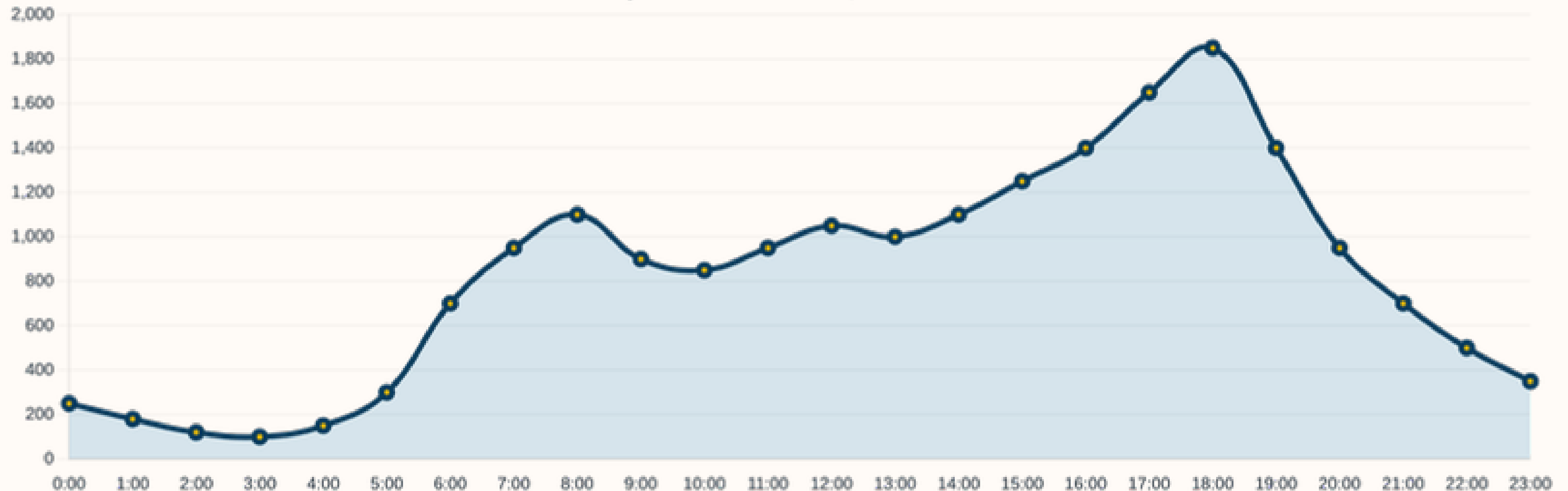


Adherence to Seat Belt Use

Temporal Analysis: Highest-Risk Hours

Peak occurrences:
6PM

Distribution of Accidents Throughout the Day



Conclusions and Next Steps

Summary of Results

⚠️ **Class Imbalance:** The low frequency of fatal cases (0.5%) requires special attention to avoid biased models.

🔑 **Critical Variables:** Male sex, peak hour (6 PM), and absence of a seatbelt are the main risk indicators.

🗄️ **Data Quality:** The 2020 dataset is robust but requires treatment of null values in columns such as “age.”

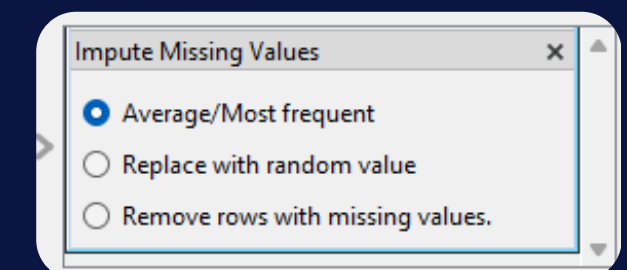
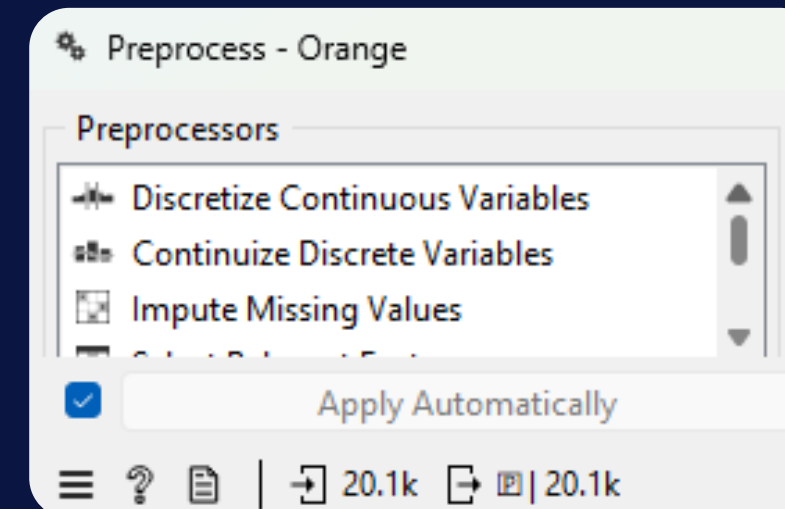
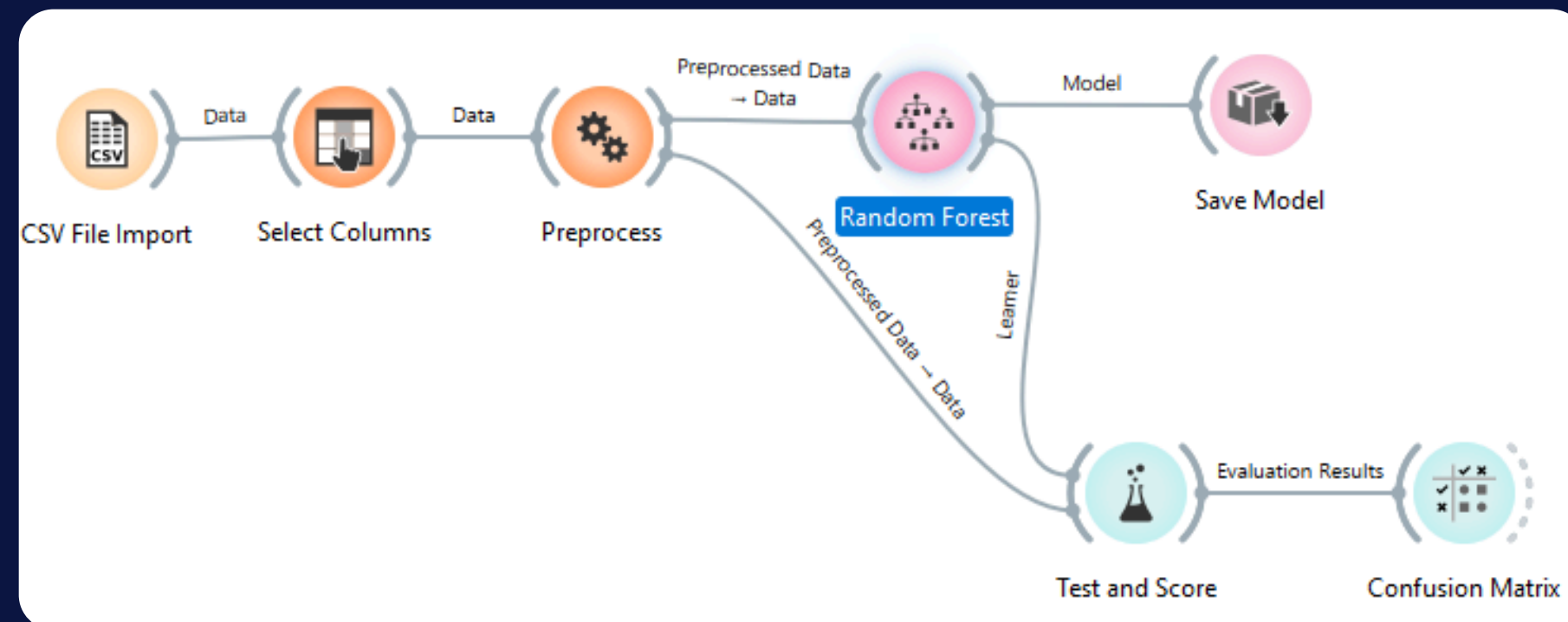


Modeling Roadmap

- 1 Preprocessing and encoding (windows-1252) of categorical variables.
- 2 Data balancing (SMOTE or Undersampling).
- 3 Algorithm training (Random Forest, 10 trees).

Focus: Maximize sensitivity (Recall) for fatal cases.

- 4 Evaluation via Confusion Matrix and F1-Score.



Modeling Roadmap

4

Evaluation via Confusion Matrix and F1-Score.

Test and Score - Orange

☒ Cross validation

Number of folds: 5

☒ Stratified

☐ Cross validation by feature

☐ Random sampling

Repeat train/test: 10

Training set size: 66 %

☒ Stratified

Evaluation results for target (None, show average over classes)

Model	Train	Test	AUC	CA	F1	Prec	Recall	MCC
Random Forest	0.836	0.136	0.894	0.862	0.861	0.860	0.862	0.722

Confusion Matrix

		Predicted			
		Crítico	Leve Moderado		Σ
Actual	Crítico	0	13	59	72
	Leve	1	7548	1251	8800
	Moderado	4	1441	9752	11197
Σ		5	9002	11062	20069

● Sistema Ativo - IA em Operação

DETECÇÃO DE *SEVERIDADE* EM ACIDENTES

Inteligência Artificial aplicada à análise em tempo real de ocorrências viárias. Classifique, priorize e acelere o atendimento de emergência.

Iniciar Análise →

Ver Como Funciona

98%

Precisão

2s

Tempo de Resposta

4x

Mais Rápido

● ALERTA CRÍTICO

Rodovia BR-101
Km 234 - Norte

CRÍTICO



● MODERADO

Av. Boa Viagem

Grupamento - Recife

MODERADO

Apelar para melhor



**THANK YOU
FOR YOUR
ATTENTION**